

Produit : Electronic / Measuring devices

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INDIA

Product's definition

Category

In which category the product is classified ?

It is classified in the category of 'Computer Hardware, Peripherals and identification Cards'.

It is regulated under IS 13252-1 (2010) : Information Technology Equipment – Safety, Part 1: General Requirements [LITD 16: Computer Hardware, Peripherals and identification Cards]

These products are governed by WPC certification norms.

Definition

What is the exact definition of the product ?

And if it exists, what is the interpretation of the authorities regarding it ?

No such Exact Definition.

Not Applicable.

Comments

Comments

IS 13252-1 (2010) : Information Technology Equipment – Safety, Part 1: General Requirements [LITD 16: Computer Hardware, Peripherals and identification Cards]

NATIONAL FOREWORD

This Indian Standard (Part 1) (Second Revision) which is identical with IEC 60950-1 : 2005 'Information technology equipment — Safety — Part 1: General requirements' issued by the International Electrotechnical Commission (IEC) was adopted by the Bureau of Indian Standards on the recommendation of the Computer Hardware, Peripherals and Identification Cards Sectional Committee and approval of the Electronics and Information Technology Division Council.

This standard was originally published in 1992 and was identical to IEC 950 : 1986. The standard was revised in 2003 based on ISO/IEC 60950-1 : 2001. This standard is now again being revised to align it with the latest IEC Publication IEC 60950-1 : 2005.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
- b) Comma (,) has been used as a decimal marker in the International Standard while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their respective places are listed below along with their degree of equivalence for the editions indicated:

International Standard	Corresponding Indian Standard	Degree of Equivalence
IEC 60065: 2001 Audio, video and similar electronic apparatus — Safety Requirements	IS 616 : 2003 Audio, video and similar electronic apparatus — Safety requirements (third revision)	Identical
IEC 60068-2-78 : 2001 Environmental testing — Part 2-78: Tests — Test Cab: Damp heat, steady state	IS 9000 (Part 4) : 2008 Basic environmental testing procedures for electronic and electrical items: Part 4 Damp heat (steady state) (first revision)	Identical
IEC 60085 : 2004 Electrical insulation — Thermal classification	IS 1271 : 1985 Thermal evaluation and classification of electrical insulation (first revision)	Technically Equivalent
IEC 60112 : 2003 Method for determination of the proof and the comparative tracking indices of insulating materials	IS 2824 : 2007 Method for determination of the proof and the comparative tracking indices of insulating materials (second revision)	Identical
IEC 60227 (All Parts) Polyvinyl chloride insulated cables of rated voltages up to and including 450/750V	IS 694 : 1990 PVC insulated cables for working voltages up to and including 100 V (third revision)	Technically Equivalent
IEC 60309 (Various Parts) Plugs, socket-outlets and couplers for industrial purposes	IS/IEC 60309 (Part 1) : 2002 Plugs, socket-outlets and couplers for industrial purposes: Part 1 General requirements (first revision)	Identical
	IS/IEC 60309 (Part 2) : 2002 Plugs, socket-outlets and couplers for industrial purposes: Part 2 Dimensional interchangeability requirements for pin and contact-tube accessories (first revision)	
IEC 60317 (Various Parts) Specifications for particular types of winding wires	IS 13730 (Various Parts) Specifications for particular types of winding wires	Identical

IEC 60320 (All Parts) Appliance couplers for household and similar general purposes	IS/IEC 60320 (All Parts) Appliance couplers for household and similar general purposes	do
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IEC 60384-14 : 1993 Fixed capacitors for use in electronic equipment — Part 14: Sectional specification: Fixed capacitors for electromagnetic interference suppression and connection to the supply mains	IS QC 302400 : 1994 Fixed capacitors for use in electronic equipment: Sectional specification for fixed capacitors for electromagnetic interference suppression and connection to the supply mains	do
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IEC 60664-1 : 2002 Insulation coordination for equipment within low-voltage systems — Part 1: Principles, requirements and tests	IS 15382 (Part 1) : 2003 Insulation coordination for equipment within low-voltage systems: Part 1 Principle requirements and tests	do
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IEC 60695-11-5 : 2004 Fire hazard testing — Part 11-5: Test flames — Needle-flame test method — Apparatus, confirmatory test arrangement and guidance	IS 11000 (Part 2/Sec 2) : 2008 Fire hazard testing: Part 2 Test methods, Section 2 Needle flame test method — Apparatus, confirmatory test arrangement and guidance (first revision)	do
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IEC 60730-1 : 1999 Automatic electrical controls for household and similar use — Part 1: General requirements	IS 4165 : 1991 Thermostats for general purpose electric ovens (first revision)	Technically Equivalent
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IEC 60747-5 : 1992 Discrete semiconductor devices — Part 5: Optoelectronic devices — Photocouplers, optocouplers	IS 14901 (Part 5) : 2004 Semiconductor devices — Discrete devices and integrated circuits: Part 5 Optoelectronic devices	Identical
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IEC 60825-2 : 1993 Safety of laser products — Part 2: Safety of optical fibre communication systems	IS 14624 (Part 2) : 1998 Safety of laser products: Part 2 Safety of optical fibre communication system	do
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IEC 60947-1 : 2004 Low voltage switchgear and control gear — Part 1: General rules	IS/IEC 60947-1 : 2004 Low voltage switchgear and control gear: Part 1 General rules	do
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IEC 60990 : 1999 Methods of measurement of touch current and protective conductor current	IS/IEC 60990 : 1999 Methods of measurement of touch current and protective conductor current	do
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IEC 61058-1 : 2000 Switches for appliances — Part 1: General requirements	IS/IEC 61058-1 : 2000 Switches for appliances: Part 1 General requirements	do
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ISO 178 : 1993 Plastics — Determination of flexural properties	IS 13360 (Part 5/Sec 7) : 1996 Plastics — Methods of testing: Part 5 Mechanical properties, Section 7 Determination of flexural properties	do
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ISO 179 (Various Parts) Plastics — Determination of Charpy impact properties	IS 13360 (Part 5/Sec 5) : 1996 Plastics — Methods of testing: Part 5 Mechanical properties, Section 5 Determination of Charpy impact strength	do
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ISO 180 : 1993 Plastics — Determination of Izod impact strength	IS 13360 (Part 5/Sec 4) : 1996 Plastics — Methods of testing: Part 5 Mechanical properties, Section 4 Determination of izod impact strength	do
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ISO 261 : 1998 ISO General purpose metric screw threads — General plan	IS 4218 (Part 2) : 2001 ISO General purpose metric screw threads: Part 2 General plan
ISO 262 : 1998 ISO General purpose metric screw threads — Selected sizes for screws, bolts and nuts	IS 4218 (Part 4) : 2007 ISO General purpose metric screw threads: Part 4 Selected sizes for screws, bolts and nuts (third revision)
ISO 527 (Various Parts) Plastics — Determination of tensile properties	IS 13360 (Part 5/Sec 1) : 1996 Plastics — Methods of testing: Part 5 Mechanical properties, Section 1 Determination of tensile properties — General principles
	IS 13360 (Part 5/Sec 2) : 1996 Plastics — Methods of testing: Part 5 mechanical properties, Section 2 Determination of tensile properties — Test conditions for moulding and extrusion plastics
	IS 13360 (Part 5/Sec 25) : 2004 Plastics — Methods of testing: Part 5 mechanical properties, Section 25 Determination of tensile properties — Test conditions for isotropic and orthotropic fibre — Reinforced plastic composites
	IS 13360 (Part 5/Sec 26) : 2004 Plastics — Methods of testing: Part 5 mechanical properties, Section 26 Determination of tensile properties — Test conditions for unidirectional fibre — Reinforced plastic composite
ISO 9773 : 1998 Plastics — Determination of burning behaviour of thin flexible vertical specimens in contact with a small-flame ignition source	IS 13360 (Part 6/Sec 23) : 2006 Plastics — Methods of testing: Part 6 Thermal properties, Section 23 Determination of burning behaviour of thin flexible vertical specimens in contact with small-flame ignition source Identical

The technical committee has reviewed the provisions of following International Standards referred in this adopted standard and has decided that they are acceptable for use in conjunction with this standard. For undated references, the latest edition of the referenced document applies, including any corrigenda and amendment.

International Standard	Title
IEC 60073	Basic and safety principles for man-machine interface, marking and identification — Coding principles for indicators and actuators
IEC 60083	Plugs and socket-outlets for domestic and similar general use standardized in member countries of IEC

IEC 60216-4-1	Guide for the determination of thermal endurance properties of electrical insulating materials — Part 4: Ageing ovens — Section 1: Single-chamber ovens
IEC 60364-1 : 2001	Electrical installations of buildings: Part 1 Fundamental principles, assessment of general characteristics, definitions
IEC 60245 (All Parts)	Rubber insulated cables — Rated voltages up to and including 450/750 V
IEC 60364-1 : 2001	Electrical installations of buildings: Part 1 Fundamental principles, assessment of general characteristics, definitions
IEC 60417-DB : 2002	Graphical symbols for use on equipment
IEC 60695-2-11	Fire hazard testing — Part 2-11: Glowing/hot-wire based test methods — Glowwire flammability test method for end-products
IEC 60695-2-20	Fire hazard testing — Part 2-20: Glowing/hot-wire based test methods — Hotwire coil ignitability — Apparatus, test method and guidance
IEC 60695-10-2	Fire hazard testing — Part 10-2: Abnormal heat — Ball pressure test
IEC 60695-11-3	Fire hazard testing — Part 11-3: Test flames — 500 W flames — Apparatus and confirmational test methods
IEC 60695-11-4	Fire hazard testing — Part 11-4: Test flames — 50 W flames — Apparatus and confirmational test methods
IEC 60695-11-10	Fire hazard testing — Part 11-10: Test flames — 50 W horizontal and vertical flame test methods
IEC 60695-11-20	Fire hazard testing — Part 11-20: Test flames — 500 W flame test methods
IEC 60825-1	Safety of laser products — Part 1: Equipment classification, requirements and user's guide
IEC 60825-9	Safety of laser products — Part 9: Compilation of maximum permissible exposure to incoherent optical radiation
IEC 60825-12	Safety of laser products — Part 1: Equipment classification, requirements and user's guide
IEC 60851-3 : 1996	Winding wires — Test methods — Part 3: Mechanical properties
IEC 60851-5 : 1996	Winding wires — Test methods — Part 5: Electrical properties
IEC 60851-6 : 1996	Winding wires — Test methods — Part 6: Thermal properties

IEC 60885-1 : 1987	Electrical test methods for electric cables — Part 1: Electrical tests for cables, cords and wires for voltages up to and including 450/750 V
IEC 60906-1	IEC system of plugs and socket-outlets for household and similar purposes — Part 1: Plugs and socket-outlets 16A 250 V a.c
IEC 60906-2	IEC system of plugs and socket-outlets for household and similar purposes — Part 2: Plugs and socket-outlets 15A 125 V a.c.
IEC 61051-2 : 1991	Varistors for use in electronic equipment — Part 2: Sectional specification for surge suppression varistors
ISO 3864 (All Parts)	Graphical symbols — Safety colours and safety signs
ISO 4892-1	Plastics — Methods of exposure to laboratory light sources — Part 1: General guidance
ISO 4892-2	Plastics — Methods of exposure to laboratory light sources — Part 2: Xenon-arc sources
ISO 4892-4	Plastics — Methods of exposure to laboratory light sources — Part 4: Openflame carbon-arc lamps
ISO 7000-DB : 2004	Graphical symbols for use on equipment — Index and synopsis
ISO 8256	Plastics — Determination of tensile-impact strength
ISO 9772	Cellular plastics — Determination of horizontal burning characteristics of small specimens subjected to a small flame
ITU-T	Recommendation K. 44, Resistibility tests for telecommunication equipment exposed to over voltages and over currents — Basic recommendation

Only the English language text of the International Standard has been retained while adopting it as an Indian Standard, and as such the page numbers given here are not the same as in the IEC Publication.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Labelling

Language

In which language must the information on the product or on the packaging be indicated? If English is sufficient please indicate NA.

Note: Generally All the particulars and Labelling are made in English Language.

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

9. Manner in which declaration shall be made

4) The particulars of the declarations required to be specified under this rule on a package shall either be in Hindi in Devanagiri script or in English; Provided...

Mandatory labelling on product

What are the mandatory labelling on the product ?

Not Specified.

In order to sell your product in B to C, the brand Decathlon has to be marked in a visible and a permanent way. Stickers are not accepted. It has to appear on the product, not only the packaging.

Mandatory labelling on packaging

What are the mandatory labelling on the packaging ?

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

4. Regulation for pre-packing and sale etc., of commodities in packaged form.-... no person shall pre-pack or cause or permit to be pre-packed any commodity for sale, distribution or delivery unless the package in which the commodity is pre-packed bears thereon, or on a label is securely affixed thereto, such declarations as are required to be made under these rules.

6. Declarations to be made on every package- All product packages, to which the Packaging Rules apply, are required to bear certain declarations on their principal display panel.

1.

1. The name and address of the manufacturer, or where the manufacturer is not packer, the name and address of the manufacturer and packer and for any imported package the name and address of the importer shall be mentioned.
2. The common or generic names of the commodity contained in the package and in case of packages with more than one product, the name and number or quantity of each product shall be mentioned on the package.
3. The net quantity, in terms of the standard unit of weight or measure, of the commodity contained in the package or where the commodity is packed or sold by number, the number of the commodity contained in the package shall be mentioned.
4. The month and year in which the commodity is manufactured or pre-packed or imported shall be mentioned in the package...
5. Retail sale price – The retail sale price to be mentioned on the package must necessarily be the maximum retail price inclusive of all taxes. The price in Indian rupees and paise should be rounded off to the nearest rupee or 50 paise.
6. Country - It is now mandatory to declare on the product package, the name of the country of origin or manufacture or assembly in case of imported products.
7. No dual MRP - The Amendment prohibits provision of dual maximum retail price (MRP) on the products. Accordingly, no manufacturer, packer or importer is permitted to declare different MRP's on identical pre-packaged commodities unless otherwise permitted.
8. Every package shall bear the name, address, telephone number, E-mail address, if applicable, of the person who can be or the office which can be, contacted, in case of consumer complaints.

9. It shall not be permissible to affix individual stickers on the package for altering or making declaration required under these rules: Provided that for reducing the Maximum Retail Price (MRP), a sticker with the revised lower MRP (inclusive of all taxes) may be affixed and the same shall not cover the MRP declaration made by the manufacturer or the packer, as the case may be, on the label of the package.
10. It shall be permissible to use stickers for making any declaration other than the declaration required to be made under these rules.
11. Where a commodity consists of a number of components and these components are to be packed in two or more units, for sale as a single commodity, the declaration required to be made under sub-rule (1) shall appear on the main package and such package shall also carry information about the other accompanying packages or such declaration may be given on individual packages and intimation to that effect may be given on the main package and if the components are sold as spare parts, all declarations shall be given on each package.

7. Principal display panel its area, size and letter etc :

(2) (i) TABLE-II

Minimum Height of Numeral

Serial Number	Net quantity in weight / volume		Minimum Height in mm
		Normal Case	When blown, formed, molded, embossed or perforated on container
1	Upto 100 cm sq.	1	2
2	Above 100 cm sq. and upto 5002 cm sq.		4
3	Above 500 cm sq. and upto 2500 cm sq.	4	6
4	Above 2500 cm sq.	6	6

(3) The height of letters in the declaration shall not be less than 1 mm height and when blown, formed, molded, embossed or perforated, the height of letters shall not be less than 2 mm.

Provided that the width of the letter or numeral shall not be less than one third of its height, except in the case of numeral '1' and letters (i), (l) and (I);

(4) the sub-rule (1)-(3) shall not apply if the information to be specified on such package under this rule is also required to be given by or under any other law for the time being in force.

8. Declaration Where to appear:- (1) Every declaration required to be made under these rules shall appear on the principal display panel:

Provided that the area surrounding the quantity declaration shall be free from printed information.

1. Above and below by a space equal to at least the height of the numeral in the declaration, and
2. To the left and right by a space at least twice the height of the numeral in the declaration.

9. Manner in which declaration shall be made;- (1) Every declaration which is required to be made on a package under these rules shall be-

(a) Legible and prominent;

(b) numerals of the retail sale price and net quantity declaration shall be printed, printed or inscribed on the package in a colour that contrasts conspicuously with the background of the label....

(3) Where a package is provided with an outside container or wrapper such container or wrapper shall also contain all the declarations which are required to appear on the package except where such container or wrapper itself is transparent and the declarations on the package itself are easily readable through such outside container or wrapper; provided...

Mandatory labelling on the instructions manual

What are the mandatory labelling on the instructions manual (if one is required) ?
In which format, do we have to deliver the instruction manual to our customers ?

Not Applicable

Warnings - General ones

Are general warnings required ? If yes, which ones ?

Not Applicable

Warnings - Specific ones

Are specific warnings required ? If yes which ones ?

Not Applicable

Target Age

Is there any target age applicable? If so, does it have to be indicated and in which form?

Not Applicable

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

20. Action to be taken on completion of inspection of packages at the premises of the manufacturer or the packer:- (1) If it appears from the report referred to in sub-rule (3) of rule 19 that,-

(a) the statistical average of the net quantity contained in the packages drawn as samples under that rule is lesser than the quantity declared on the packages or on the labels affixed thereto, or any such package shows an error in deficiency greater than the maximum permissible error, or

(b) any such package does not bear thereon or on label affixed thereto the declarations to be made under these rules, the Director, Controller or any Legal Metrology Officer shall take action as given below, namely:-

(i) seize the packages drawn by him as samples and shall take adequate steps for the safe custody of seized packages until they are produced in the appropriate court as evidence;

(ii) based on the evidence initiate action for violations of the provisions of the Act and these rules:

Provided that no such action shall be taken if fresh tests are carried out under sub-rule (4) of rule 19, but if after such fresh tests any error or omission as is referred to in this sub-rule is detected, the Director, Controller or any Legal Metrology Officer shall take appropriate action as specified in this sub-rule in accordance with the provisions of the Act against the manufacturer or, as the case may be, the packer.

(3) The disposal of the seized packages shall be made in accordance with the provisions of the Code of Criminal Procedure, 1973 (2 of 1974).

Chapter VII GENERAL

32. Penalty for contravention of Rules.-(1) Whoever contravenes the provisions of rules 27 to 31, he shall be punished with fine of four thousand rupees.

(2) Whoever contravenes any other provision of these rules, for the contravention of which no punishment has been provided either in the Act or in the rules, he shall be punished with fine of two thousand rupees.

Comments

Comments

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

18. Provisions relating to wholesale dealer and retail dealers: (Refer to authority attached)

19. Inspection of quantity and error in packages at the premises of the manufacturer or packer:- (Refer to authority attached)

21. Inspection of quantity and error in packages at the premises of the wholesale dealer or retail dealer :- (Refer to authority attached)

23. Deceptive packages to be repacked or in default to be seized: (Refer to authority attached)

Chapter IV EXEMPTIONS

26. Exemption in respect of certain packages: - Nothing contained in these rules shall apply to any package containing a commodity if-

(a) the net weight or measurement of the commodity is ten gram or ten milliliter or less, if sold by weight or measurement.(**)

Chapter VI REGISTRATION OF MANUFACTURERS, PACKERS AND IMPORTERS

27. Registration of manufacturers, Packers and importers :(Refer to authority attached)

28. Registration of shorter address permissible :.. (Refer to authority attached)

29. Registration of manufacturers and packers, etc.: .. (Refer to authority attached)

Chapter VII GENERAL

31. (1) Any advertisement mentioning the retail ale price of the pre-packaged commodity shall contain a declaration as to the net quantity or number of the commodity contained in the package.

(2) The font size of the net quantity in the advertisement shall be same as that of retail sale price.

30. Compilation of lists of manufacturers or packers and their circulation...

33. Repeal and savings ...

Environment

Recycling - Eco taxes

Which regulations concerning ecodesign and recycling apply to this product? Are there any ecotaxes applicable to this category of product?

Not Applicable

Labelling

What are the mandatory labellings to put on the product or on the packaging regarding the environmental rules ?

Not Applicable

Technical rules / Standards

General safety requirements

What are the general safety requirements ? Does a risk assessment have to be conducted ? If yes, does it concern specific risk ?

Refer to question of “Mandatory Standards” in the Technical Section.

Mandatory or voluntary standards

Are there some mandatory or voluntary standards applicable ? If yes, which ones ?

Please indicate the reference of the applicable standards

IS 13252-1 (2010) : Information Technology Equipment – Safety, Part 1: General Requirements [LITD 16: Computer Hardware, Peripherals and identification Cards]

INTRODUCTION

0 Principles of safety

The following principles have been adopted by technical committee 108 in the development of this standard.

These principles do not cover performance or functional characteristics of equipment.

Words printed in SMALL CAPITALS are terms that are defined in 1.2 of this standard.

0.1 General principles of safety

It is essential that designers understand the underlying principles of safety requirements in order that they can engineer safe equipment.

These principles are not an alternative to the detailed requirements of this standard, but are intended to provide designers with an appreciation of the basis of these requirements. Where the equipment involves technologies and materials or methods of construction not specifically covered, the design of the equipment should provide a level of safety not less than those described in these principles of safety.

Designers shall take into account not only normal operating conditions of the equipment but also likely fault conditions, consequential faults, foreseeable misuse and external influences such as temperature, altitude, pollution, moisture, over voltages on the MAINS SUPPLY and over voltages on a TELECOMMUNICATION NETWORK or a CABLE DISTRIBUTION SYSTEM. Dimensioning of insulation spacings should take account of possible reductions by manufacturing tolerances, or where deformation could occur due to handling, shock and vibration likely to be encountered during manufacture, transport and normal use.

The following priorities should be observed in determining what design measures to adopt: where possible, specify design criteria that will eliminate, reduce or guard against hazards; where the above is not practicable because the functioning of the equipment would be impaired, specify the use of protective means independent of the equipment, such as personal protective equipment (which is not specified in this standard); where neither of the above measures is practicable, or in addition to those measures, specify the provision of markings and instructions regarding the residual risks. There are two types of persons whose safety needs to be considered, USERS (or OPERATORS) and SERVICE PERSONS.

USER is the term applied to all persons other than SERVICE PERSONS. Requirements for protection should assume that USERS are not trained to identify hazards, but will not intentionally create a hazardous situation. Consequently, the requirements will provide protection for cleaners and casual visitors as well as the assigned USERS. In general, USERS should not have access to hazardous parts, and to this end, such parts should only be in SERVICE ACCESS AREAS or in equipment located in RESTRICTED ACCESS LOCATIONS.

When USERS are admitted to RESTRICTED ACCESS LOCATIONS they shall be suitably instructed.

SERVICE PERSONS are expected to use their training and skill to avoid possible injury to themselves and others due to obvious hazards that exist in SERVICE ACCESS AREAS of the equipment or on equipment located in RESTRICTED ACCESS LOCATIONS. However, SERVICE PERSONS should be protected against unexpected hazards. This can be done by, for example, locating parts that need to be accessible for servicing away from electrical and mechanical hazards, providing shields to avoid accidental contact with hazardous parts, and providing labels or instructions to warn personnel about any residual risk.

Information about potential hazards can be marked on the equipment or provided with the equipment, depending on the likelihood and severity of injury, or made available for SERVICE PERSONS. In general, USERS shall not be exposed to hazards likely to cause injury, and information provided for USERS should primarily aim at avoiding misuse and situations likely to create hazards, such as connection to the wrong power source and replacement of fuses by incorrect types.

MOVABLE EQUIPMENT is considered to present a slightly increased risk of shock, due to possible extra strain on the supply cord leading to rupture of the earthing conductor. With HAND-HELD EQUIPMENT, this risk is increased; wear on the cord is more likely, and further hazards could arise if the units were dropped. TRANSPORTABLE EQUIPMENT introduces a further factor because it can be used and carried in any orientation; if a small metallic object enters an opening in the ENCLOSURE it can move around inside the equipment, possibly creating a hazard.

0.2 Hazards

Application of a safety standard is intended to reduce the risk of injury or damage due to the following:

electric shock;

energy related hazards;

fire;

heat related hazards;

mechanical hazards;

radiation;

chemical hazards.

0.2.1 Electric shock

Electric shock is due to current passing through the human body. The resulting physiological effects depend on the value and duration of the current and the path it takes through the body. The value of the current depends on the applied voltage, the impedance of the source and the impedance of the body. The body impedance depends in turn on the area of contact, moisture in the area of contact and the applied voltage and frequency. Currents of approximately half a milliampere can cause a reaction in persons in good health and may cause injury indirectly due to involuntary reaction. Higher currents can have more direct effects, such as burn or muscle tetanization leading to inability to let go or to ventricular fibrillation.

Steady state voltages up to 42,4 V peak, or 60 V d.c., are not generally regarded as hazardous under dry conditions for an area of contact equivalent to a human hand. Bare parts that have to be touched or handled should be at earth potential or properly insulated.

Some equipment will be connected to telephone and other external networks. Some TELECOMMUNICATION NETWORKS operate with signals such as voice and ringing superimposed on a steady d.c. supply voltage; the total may exceed the values given above for steady-state voltages. It is common practice for the SERVICE PERSONS of telephone companies to handle parts of such circuits bare-handed. This has not caused serious injury, because of the use of cadenced ringing and because there are limited areas of contact with bare conductors normally handled by SERVICE PERSONS. However, the area of contact of a part accessible to the USER, and the likelihood of the part being touched, should be further limited (for example, by the shape and location of the part).

It is normal to provide two levels of protection for USERS to prevent electric shock. Therefore, the operation of equipment under normal conditions and after a single fault, including any consequential faults, should not create a shock hazard. However, provision of additional protective measures, such as protective earthing or SUPPLEMENTARY INSULATION, is not considered a substitute for, or a relief from, properly designed BASIC INSULATION.

Harm may result from:

Examples of measures to reduce risks:

Contact with bare parts normally at HAZARDOUS VOLTAGES.	Prevent USER access to parts at HAZARDOUS VOLTAGES by fixed or locked covers, SAFETY INTERLOCKS, etc. Discharge accessible capacitors that are at HAZARDOUS VOLTAGES.
Breakdown of insulation between parts normally at HAZARDOUS VOLTAGES and accessible conductive parts.	Provide BASIC INSULATION and connect the accessible conductive parts and circuits to earth so that exposure to the voltage which can develop is limited because overcurrent protection will disconnect the parts having low impedance faults within a specified time; or provide a metal screen connected to protective earth between the parts, or provide DOUBLE INSULATION or REINFORCED INSULATION between the parts, so that breakdown to the accessible part is not likely to occur.
Contact with circuits connected to TELECOMMUNICATION NETWORKS that exceed 42,4 V peak or 60 V d.c.	Limit the accessibility and area of contact of such circuits, and separate them from unearthed parts to which access is not limited.
Breakdown of USER-accessible insulation.	Insulation that is accessible to the USER should have adequate mechanical and electrical strength to reduce the likelihood of contact with HAZARDOUS VOLTAGES.
TOUCH CURRENT (leakage current) flowing from parts at HAZARDOUS VOLTAGES to accessible parts, or failure of a protective earthing connection. TOUCH CURRENT may include current due to EMC filter components connected between PRIMARY CIRCUITS and accessible parts.	Limit TOUCH CURRENT to a specified value, or provide a high integrity protective earthing connection.

0.2.2 Energy related hazards

Injury or fire may result from a short-circuit between adjacent poles of high current supplies or high capacitance circuits, causing:

- burns;
- arcing;
- ejection of molten metal.

Even circuits whose voltages are safe to touch may be hazardous in this respect.

Examples of measures to reduce risks include:

- separation;
- shielding;
- provision of SAFETY INTERLOCKS.

0.2.3 Fire

Risk of fire may result from excessive temperatures either under normal operating conditions or due to overload, component failure, insulation breakdown or loose connections. Fires originating within the equipment should not spread beyond the immediate vicinity of the source of the fire, nor cause damage to the surroundings of the equipment.

Examples of measures to reduce risks include:

- providing overcurrent protection;
- using constructional materials having appropriate flammability properties for their purpose;
- selection of parts, components and consumable materials to avoid high temperature which might cause ignition;
- limiting the quantity of combustible materials used;
- shielding or separating combustible materials from likely ignition sources;
- using ENCLOSURES or barriers to limit the spread of fire within the equipment;
- using suitable materials for ENCLOSURES so as to reduce the likelihood of fire spreading from the equipment.

0.2.4 Heat related hazards

Injury may result from high temperatures under normal operating conditions, causing:

- burns due to contact with hot accessible parts;
- degradation of insulation and of safety-critical components;
- ignition of flammable liquids.

Examples of measures to reduce risks include:

- taking steps to avoid high temperature of accessible parts;
- avoiding temperatures above the ignition point of liquids;
- provision of markings to warn USERS where access to hot parts is unavoidable.

0.2.5 Mechanical hazards

Injury may result from:

- sharp edges and corners;
- moving parts that have the potential to cause injury;
- equipment instability;
- flying particles from imploding cathode ray tubes and exploding high pressure lamps.

Examples of measures to reduce risks include:

rounding of sharp edges and corners;

guarding;

provision of SAFETY INTERLOCKS;

providing sufficient stability to free-standing equipment;

selecting cathode ray tubes and high pressure lamps that are resistant to implosion and explosion respectively;

provision of markings to warn USERS where access is unavoidable.

0.2.6 Radiation

Injury to USERS and to SERVICE PERSONS may result from some forms of radiation emitted by equipment. Examples are sonic (acoustic), radio frequency, infra-red, ultraviolet and ionizing radiation, and high intensity visible and coherent light (lasers).

Examples of measures to reduce risks include:

limiting the energy level of potential radiation sources;

screening radiation sources;

provision of SAFETY INTERLOCKS;

provision of markings to warn USERS where exposure to the radiation hazard is unavoidable.

0.2.7 Chemical hazards

Injury may result from contact with some chemicals or from inhalation of their vapours and fumes.

Examples of measures to reduce risks include:

avoiding the use of constructional and consumable materials likely to cause injury by contact or inhalation during intended and normal conditions of use;

avoiding conditions likely to cause leakage or vaporization;

provision of markings to warn USERS about the hazards.

0.3 Materials and components

Materials and components used in the construction of equipment should be so selected and arranged that they can be expected to perform in a reliable manner for the anticipated life of the equipment without creating a hazard, and would not contribute significantly to the development of a serious fire hazard. Components should be selected so that they remain within their manufacturers' ratings under normal operating conditions, and do not create a hazard under fault conditions.

PART 1 GENERAL REQUIREMENTS

1 General

1.1 Scope

1.1.1 Equipment covered by this standard This standard is applicable to mains-powered or battery-powered information technology equipment, including electrical business equipment and associated equipment, with a RATED VOLTAGE not exceeding 600 V.

This standard is also applicable to such information technology equipment:

designed for use as telecommunication terminal equipment and TELECOMMUNICATION NETWORK infrastructure equipment, regardless of the source of power;

designed and intended to be connected directly to, or used as infrastructure equipment in, a CABLE DISTRIBUTION SYSTEM, regardless of the source of power;

designed to use the AC MAINS SUPPLY as a communication transmission medium (see Clause 6, Note 4 and 7.1, Note 4).

This standard is also applicable to components and subassemblies intended for incorporation in information technology equipment. It is not expected that such components and subassemblies comply with every aspect of the standard, provided that the complete information technology equipment, incorporating such components and subassemblies, does comply.

NOTE 1 Examples of aspects with which uninstalled components and subassemblies may not comply include the marking of the power rating and access to hazardous parts.

NOTE 2 This standard may be applied to the electronic parts of equipment even if that equipment does not wholly fall within its Scope, such as large-scale air conditioning systems, fire detection systems and fire extinguishing systems. Different requirements may be necessary for some applications.

This standard specifies requirements intended to reduce risks of fire, electric shock or injury for the OPERATOR and layman who may come into contact with the equipment and, where specifically stated, for a SERVICE PERSON.

This standard is intended to reduce such risks with respect to installed equipment, whether it consists of a system of interconnected units or independent units, subject to installing, operating and maintaining the equipment in the manner prescribed by the manufacturer.

Examples of equipment that is in the scope of this standard are:

Generic product type	Specific example of generic type
banking equipment	monetary processing machines including automated teller (cash dispensing) machines (ATM)
data and text processing machines and associated equipment	data preparation equipment, data processing equipment, data storage equipment, personal computers, plotters, printers, scanners, text processing equipment, visual display units
data network equipment	bridges, data circuit terminating equipment, data terminal equipment, routers
electrical and electronic retail equipment	cash registers, point of sale terminals including associated electronic scales

electrical and electronic office machines	calculators, copying machines, dictation equipment, document shredding machines, duplicators, erasers, micrographic office equipment, motor-operated files, paper trimmers (punchers, cutting machines, separators), paper jogging machines, pencil sharpeners, staplers, typewriters
other information technology equipment	photoprinting equipment, public information terminals, multi-media equipment
postage equipment	mail processing machines, postage machines
telecommunication network infrastructure equipment	billing equipment, multiplexers, network powering equipment, network terminating equipment, radio basestations, repeaters, transmission equipment, telecommunication switching equipment
telecommunication terminal equipment	facsimile equipment, key telephone systems, modems, PABXs, pagers, telephone answering machines, telephone sets (wired and wireless)

NOTE 3 The requirements of IEC 60065 may also be used to meet safety requirements for multimedia equipment. See IEC Guide 112, Guide on the safety of multimedia equipment.

This list is not intended to be comprehensive, and equipment that is not listed is not necessarily excluded from the Scope.

Equipment complying with the relevant requirements in this standard is considered suitable for use with process control equipment, automatic test equipment and similar systems requiring information processing facilities. However, this standard does not include requirements for performance or functional characteristics of equipment.

1.1.2 Additional requirements

Requirements additional to those specified in this standard may be necessary for:

equipment intended for operation in special environments (for example, extremes of temperature; excessive dust, moisture or vibration; flammable gases; and corrosive or explosive atmospheres);

electromedical applications with physical connections to the patient;

equipment intended to be used in vehicles, on board ships or aircraft, in tropical countries, or at altitudes greater than 2 000 m;

equipment intended for use where ingress of water is possible; for guidance on such requirements and on relevant testing, see Annex T.

NOTE Attention is drawn to the fact that authorities of some countries impose additional requirements.

1.1.3 Exclusions This standard does not apply to:

power supply systems which are not an integral part of the equipment, such as motor generator sets, battery backup systems and transformers;

building installation wiring;

devices requiring no electric power.

1.2 Definitions

For the purpose of this International Standard the following definitions apply. Where the terms "voltage" and "current" are used, they imply the r.m.s. values, unless otherwise specified.

Definitions in alphabetical order of noun

(*Please refer to PDF File attached of this Indian Standard for Definitions)

1.3 General requirements

1.3.1 Application of requirements

The requirements detailed in this standard shall be applied only if safety is involved.

In order to establish whether or not safety is involved, the circuits and construction shall be carefully investigated to take into account the consequences of possible failures.

1.3.2 Equipment design and construction

Equipment shall be so designed and constructed that, under all conditions of normal use and under likely abnormal use or single fault conditions (see 1.4.14), protection is provided to reduce the risk of personal injury from electric shock and other hazards, and against spread of fire originating in the equipment.

Compliance is checked by inspection and by the relevant tests.

1.3.3 Supply voltage

Equipment shall be designed to be safe at any supply voltage to which it is intended to be connected.

Compliance is checked by inspection and by carrying out the relevant tests of this standard using a supply voltage as specified in the corresponding subclause. If the subclause does not specify the supply voltage (explicitly or by reference to 1.4.5), then the value of the RATED VOLTAGE or any value in the RATED VOLTAGE RANGE shall be used.

1.3.4 Constructions not specifically covered

Where the equipment involves technologies and materials or methods of construction not specifically covered in this standard, the equipment shall provide a level of safety not less than that generally afforded by this standard and the principles of safety contained herein.

NOTE The need for additional detailed requirements to cope with a new situation should be brought promptly to the attention of the appropriate committee.

1.3.5 Equivalent materials

Where the standard specifies a particular grade of insulation, the use of a better grade of insulation is permitted. Similarly, where the standard requires material of a particular FLAMMABILITY CLASS, the use of a better class is permitted.

1.3.6 Orientation during transport and use

Where it is clear that the orientation of the equipment is likely to have a significant effect on the application of the requirements or the results of tests, all orientations of use permitted in the installation or operating instructions shall be taken into account. For TRANSPORTABLE EQUIPMENT, all orientations of transport and use shall be taken into account.

NOTE The above may apply to 4.1, 4.2, 4.3.8, 4.5, 4.6 and 5.3.

1.3.7 Choice of criteria

Where the standard permits a choice between different criteria for compliance, or between different methods or conditions of test, the choice is specified by the manufacturer.

1.3.8 Examples mentioned in the standard

Where examples of equipment, parts, methods of construction, design techniques and faults are given in the standard, prefaced by "for example" or "such as", other examples, situations and solutions are not excluded.

1.3.9 Conductive liquids

For the electrical requirements of this standard, conductive liquids shall be treated as conductive parts.

1.4 General conditions for tests

(Please refer to PDF File attached)

1.5 Components

(Please refer to PDF File attached)

1.6 Power interface

(Please refer to PDF File attached)

1.7 Markings and instructions

(Please refer to PDF File attached)

2 Protection from hazards

2.1 Protection from electric shock and energy hazards

(Please refer to PDF File attached)

2.2 SELV circuits

(Please refer to PDF File attached)

2.3 TNV circuits

(Please refer to PDF File attached)

2.4 Limited current circuits

(Please refer to PDF File attached)

2.5 Limited power sources

(Please refer to PDF File attached)

2.6 Provisions for earthing and bonding

(Please refer to PDF File attached)

2.7 Overcurrent and earth fault protection in primary circuits

(Please refer to PDF File attached)

2.8 Safety interlocks

(Please refer to PDF File attached)

2.9 Electrical insulation

(Please refer to PDF File attached)

2.10 Clearances, creepage distances and distances through insulation

(Please refer to PDF File attached)

3 Wiring, connections and supply

3.1 General

(Please refer to PDF File attached)

3.2 Connection to a mains supply

(Please refer to PDF File attached)

3.3 Wiring terminals for connection of external conductors

(Please refer to PDF File attached)

3.4 Disconnection from the mains supply

(Please refer to PDF File attached)

3.5 Interconnection of equipment

(Please refer to PDF File attached)

4 Physical requirements

4.1 Stability Under conditions of normal

(Please refer to PDF File attached)

4.2 Mechanical strength

(Please refer to PDF File attached)

4.3 Design and construction

(Please refer to PDF File attached)

4.4 Protection against hazardous moving parts

(Please refer to PDF File attached)

4.5 Thermal requirements

(Please refer to PDF File attached)

4.6 Openings in enclosures

(Please refer to PDF File attached)

4.7 Resistance to fire

(Please refer to PDF File attached)

5 Electrical requirements and simulated abnormal conditions

5.1 Touch current and protective conductor current

(Please refer to PDF File attached)

5.2 Electric strength

(Please refer to PDF File attached)

5.3 Abnormal operating and fault conditions

(Please refer to PDF File attached)

6 Connection to telecommunication networks

(Please refer to PDF File attached)

7 Connection to cable distribution systems

(Please refer to PDF File attached)

Physical and mechanicals properties

What are the physical and mechanicals properties required ?

Refer to question of "Mandatory Standards" in the Technical Section.

Electrical aspects

What are requirements regarding electrical aspects ?

Not Applicable

Inflammability

Are there any specific requirements to ensure inflammability?

Not Applicable

Hygiene

What are the rules regarding hygiene ?

Not Applicable

Radioactivity

What are the rules regarding radioactivity ?

Not Applicable

Radiofrequency (2G,3G,4G,5G, Bluetooth, Ant+, GPS, Wifi...)

What are the rules regarding the radiofrequency?

Products making noise

What are the requirements regarding products making noise ?

Not Applicable

Chemicals substances

What are the requirements regarding chemicals substances ? Which ones are forbidden ? Limited ?
Do we have to clearly indicate all substances that are in the product, or is the indication of major

Packaging requirements

Can you please indicate if there are some requirements regarding the packaging of our products (mainly they are in paper, carton, plastic, textile) ? If yes, please explain them.

Not Applicable

Laboratory

In which laboratory can we perform the testing of the product according to the standard?

Please refer to following pdf files uploaded :

Group 1 List of BIS Recognised Laboratories as on 13-11-2019

Group 2 Lab of National Repute and Eminence, Facilities

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

Not Applicable

Comments

Comments

Certifications / Registrations

Certifications or registrations

Are there some certifications or registrations applicable to the product ?

Voluntary ISI Certificate may be procured.

Laboratory

In case of mandatory certification/ registration that we must follow please precise the name and address of any laboratories where we can conduct them

Please refer to following pdf files uploaded :

Group 1 List of BIS Recognised Laboratories as on 13-11-2019

Comments

Comments

The steps of getting ISI Certificate are as below -:

Step 1 : Confirmation of Product Quality as per concerned IS Standard

- Identification of ISI Standard Code
- Collection of the Testing Sample from applicant
- Testing of sample at our lab
- Issuing the guideline about Non-Conformity, if any
- Issue of Test Report for your reference

Step 2 : Filing of the Application Form

- Arrange the Internal Test Report of product as per test report
- Compiling of all required documents
- Remit the application fees direct to BIS in USD*
- Filling-up the Form-V
- Submission of Application Form with all enclosures
- Get the confirmation about scrutiny of documents by BIS Office

Step 3 : Remittance of Audit Fees & Travelling Expenses for Inspection

- Pay the audit fees for inspection
- Get the confirmation about BIS Inspection Team
- Organize the VISA, Ticket, Lodging & Boarding for visiting team
- Inspection team will visit to Factory for Physical Verification
- Team will verify the Factory Set-up & Quality Control Process
- Team will pick the random sample to be tested in BIS Approved Lab

Step 4 : Organize the Lab Test Report (LTR)

- Select the BIS recognized lab as per guideline of BIS Office
- Provide the sample (picked-up by team) of product to get tested
- Get the test result

Step 5 : Closure of Non-Conformity (if any) and Remit the Official Fees

- Rectify the process of Factory Unit, if inspection team points out
- Submit the proof of rectification done by unit to close non-conformity
- Pay the annual marking fees (annual license fees) to BIS
- Submit the Bank Guarantee to BIS

Step 6 : Get the ISI Marking /Registration Certificate

Conformity documents

Documents

What are the conformity documents ? Can you please list them ?

Not Applicable

For ISI Certificate, refer to comments sections of "Certifications/Registrations".

Format

Does it exist a specific format for the conformity documents ? If yes, can you please provide us an example in english ?

No Particular Format

Comments

Comments

Importation rules

General rules

What are the general rules in order to import the product ?

Generally, Importers are required to register with the DGFT to obtain an Importer Exporter Code Number (IEC) issued against their Permanent Account Number (PAN), before engaging in EXIM activities. After an IEC has been obtained, the source of items for import must be identified and declared.

IMPORT PROCEDURE

Step1 : File Bill of Entry with Business Identification Number (BIN)

Step 2: Determine rate of duty for clearance from warehouse

Step 3: File requisite documents with customs department

Step 4: Submit import report/ manifest

Step 5: Receive permission to import goods

Documents

What are the required documents ?

Mandatory Documents:

1. Importer Exporter Code issued from the DGFT
2. Bill of Entry
3. Examination Order generated by the EDI system of Customs

Optional Documents depending upon the type of Product being Imported:

1. Certificate of Origin issued by Authorized Person/ Agency at the place of manufacturing/ processing etc. of the food consignment. Certificate of Origin shall contain information on Country of Origin etc. Hence if the consignor is from a different country;
2. Stuffing list, Packing List;
3. Commercial invoice as mentioned in the Bill of Entry (BoE);
4. Bill of Lading as mentioned in the Bill of Entry (BoE) for sea consignment;
5. Air Way Bill as mentioned in the Bill of Entry (BoE) for air consignment;

Time period

In which time period may we obtain the right to import ?

It generally takes 45 to 60 days for completing the overall procedure from import clearance to Product reaching store shelf.

Administration

What is the administration in charge of importations ? Name and address ?

- Directorate General of Foreign Trade
- Central Board of Excise and Customs

Comments

Comments

Inspection of the product

Inspection method

Does it exist an inspection procedure applicable to the product ? If yes, which administration is in charge of ? Can you please describe the procedure ?

Refer to question of "Mandatory Standards" in the Technical Section.

Commercialization rules

Notification / Declaration

What are the general rules in order to commercialize a product ?
Is there a notification / declaration to do ?

Refer to question of "Mandatory Standards" in the Technical Section.

Documents

What are the required documents ?

Trade license will have to be acquired which is governed by Municipality Act- which will location specific. Notification is location specific.

Delay

What is the delay to obtain the right to commercialize ?

General documents will be required-in relation to establishment of business.

Display in store

Is there some specific rules on the way to display the product in store ?

Not Applicable

Display on internet

Is there some specific mentions to display the product on internet ?

Information to consumers

Do some specific information to have be given to consumers when the product is placed on the market ?

Not Applicable

Comments

Comments

Department of Legal Metrology which comes under the Department of Consumer Affairs.

Administration

What is the administration in charge of commercialization ? Name and address ?

Not Applicable